

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$8x^2 - 40 = 32$

What is the positive solution to the given equation?

Answer

- A. 3
- B. 4
- C. 9
- D. 72

Correct Answer: A

Rationale

Choice A is correct. Adding 40 to both sides of the given equation yields $8x^2 = 72$. Dividing both sides of this equation by 8 yields $x^2 = 9$. Taking the square root of both sides of this equation yields $x = 3$ or $x = -3$. Therefore, the positive solution to the given equation is 3.

Choice B is incorrect. This is the solution to the equation $8x = 32$, not the given equation.

Choice C is incorrect. This is the solution to the equation $8x - 40 = 32$, not the given equation.

Choice D is incorrect. This is the solution to the equation $x - 40 = 32$, not the given equation.